

JEE MAIN REVISION MASTER: HYPERBOLA SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 95 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Asymptotes, Rectangular Hyperbola & Conjugate	95 PYQs	Hyperbola
Moderate	Rectangular Hyperbola & Asymptotes	3 PYQs	Asymptotes & Geometry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Asymptotes, Rectangular Hyperbola & Conjugate	$e = \sqrt{1 + \frac{b^2}{a^2}}, \quad \text{LR: } y = mx \pm \sqrt{a^2 m^2 - b^2}, \quad \text{Tangent: } y = mx \pm \sqrt{a^2 m^2 + b^2}, \quad \text{Rectangular: } xy = c^2 \implies e = \sqrt{2}$
Rectangular Hyperbola & Asymptotes	$\text{Asymptotes: } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \implies y = \pm \frac{b}{a}x, \quad \text{Rectangular: } a = b \iff e = \sqrt{2}, \quad xy = c^2 \implies (ct, c/t)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Asymptotes, Rectangular Hyperbola & Conjugate (Very High Yield — 95 PYQs)

Core Principle: Focal distance difference property $|SP - S'P| = 2a$, eccentricity $e > 1$, asymptotes $y = \pm (b/a)x$, and rectangular hyperbola.

■ **Caution / JEE Pitfall:** For the conjugate hyperbola $-(x^2/a^2) + (y^2/b^2) = 1$, eccentricity is $e' = \sqrt{1 + a^2/b^2}$, and $(1/e^2) + (1/e'^2) = 1$.

■ **Speed Trick:** The eccentricity of ANY rectangular hyperbola ($a = b$ or $xy = c^2$) is always uniquely equal to $\sqrt{2}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm \frac{4\sqrt{6}}{3}$..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "Let $H: \frac{x^2}{2a^2} - \frac{y^2}{2b^2} = 1$ be a hyperbola such that the distance between its foci is 6 and the distance between its d..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Let the domain of the function $f(x) = \log_3 \log_5 \log_7 (9x - x^2 - 13)$ be the interval (m, n) . Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$..." [Ans: N/A]

3.2 Rectangular Hyperbola & Asymptotes (Moderate Yield — 3 PYQs)

Core Principle: Equations and properties of asymptotes, angle between asymptotes $2 \tan^{-1}(b/a)$, and parametric properties of $xy = c^2$.

■ **Caution / JEE Pitfall:** The asymptotes of a hyperbola pass through its CENTER and are parallel to the conjugate hyperbola's asymptotes.

■ **Speed Trick:** Tangent to rectangular hyperbola $xy = c^2$ at $(ct, c/t)$ is $rac{xt}{c} + \frac{yt}{c} = 2c$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "Let O be the origin, and P and Q be two points on the rectangular hyperbola $xy=12$ such that the midpoint of the line segment PQ is $(\frac{1}{2}, \frac{1}{2})$." **[Ans: N/A]**
- **[2023 Evening Shift] Q#3 (Numerical):** "Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is reciprocal to that of the hyperbola $2x^2 - 2y^2 = 1$. If..." **[Ans: 2]**
- **[2022 Evening Shift] Q#4 (MCQ):** "Let $a > 0, b > 0$. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm \frac{4\sqrt{6}}{3}$. Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E , and whose length of latus rectum is equal to the length of minor axis of E . Then the distance between the foci of H is :

- (A) $\frac{4\sqrt{2}}{\sqrt{7}}$
- (B) $\frac{4\sqrt{2}}{7}$
- (C) $\frac{4\sqrt{7}}{7}$
- (D) $(8/7)$ **[Correct]**

Correct Answer: D

Solution Analysis: For the ellipse E , let its equation be $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, A > B$ since its major axis is along the x -axis because the directrices are of the form $x = \text{constant}$. Given eccentricity, $e = \frac{\sqrt{3}}{2}$. For an ellipse with major axis along the x -axis, the directrices are $x = \pm (A/e)$. We are given $x = \pm \frac{4\sqrt{6}}{3}$. So, $(A/e) = \frac{4\sqrt{6}}{3}$. Substituting $e = \frac{\sqrt{3}}{2}$, $A = \frac{\sqrt{3}}{2} \cdot \frac{4\sqrt{6}}{3} = \frac{4\sqrt{18}}{6} = \frac{4 \cdot 3\sqrt{2}}{6} = 2\sqrt{2}$. Thus the semi-major axis of ellipse E is $2\sqrt{2}$.